

PRAKTIKUM BASIS DATA

MODUL 13

NoSQL_Key_Value_Database



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JURNAL

1. No 1

```
> db.toko.insertOne({
  name: "Cia Mart",
  category: "Minimarket",
  city: "Purwokerto",
  address: "Jl. Soedirman No. 45, Purwokerto",
  products: [
    {
      name: "Beras Ramos 5kg",
      price: 78000,
      stock: 20
    },
    {
      name: "Minyak Goreng 2L",
      price: 38000,
      stock: 15
    }
  ]
});
< {
  acknowledged: true,
  insertedId: ObjectId('6a250c73fb49ecee08a5874')
}
```

2. No 2

```
> db.toko.find({ city: "Jakarta" })
< {
  _id: ObjectId('6a250c55fb49eceeaa08a5870'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
{
  _id: ObjectId('6a250c55fb49eceeaa08a5872'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
```

```
}
{
  _id: ObjectId('6a250c55fb49ecee08a5872'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
```

3. No 3

```
> db.toko.find({
  "products.price": { $gt: 3000000 }
})
< {
  _id: ObjectId('6a250c55fb49ecee08a5870'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

```
}  
{  
  _id: ObjectId('6a250c55fb49eceeaa08a5871'),  
  name: 'Toko Maju',  
  category: 'Elektronik',  
  city: 'Bandung',  
  address: {  
    street: 'Jl. Asia Afrika',  
    zipcode: '40111'  
  },  
  products: [  
    {  
      item: 'HP',  
      price: 1500000,  
      stock: 20  
    },  
    {  
      item: 'Laptop',  
      price: 8000000,  
      stock: 3  
    }  
  ]  
}
```

```
{
  _id: ObjectId('6a250c55fb49ecee08a5872'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
```

```
_id: ObjectId('6a250c55fb49eceeaa08a5873'),
name: 'Toko Sudinotech',
category: 'Elektronik',
city: 'Surabaya',
address: {
  street: 'Jl. Sudirman',
  zipcode: '10220'
},
products: [
  {
    item: 'RTX 55000',
    price: 8500000,
    stock: 20
  },
  {
    item: 'Set PC Gaming Radeon',
    price: 12000000,
    stock: 9
  }
]
```

4. No 4

```
db.toko.updateOne(
  { name: "Toko Berkah" },
  {
    $set: {
      category: "Aksesoris"
    }
  }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

5. No 5

```
> db.toko.aggregate([
  {
    $group: {
      _id: "$city",
      jumlah_toko: { $sum: 1 }
    }
  },
  {
    $sort: {
      jumlah_toko: -1
    }
  }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Purwokerto',
  jumlah_toko: 1
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
}
```

6. No 6

```

> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$category",
      rata_rata_harga: {
        $avg: "$products.price"
      }
    }
  },
  {
    $sort: {
      rata_rata_harga: -1
    }
  }
])
< {
  _id: 'Elektronik',
  rata_rata_harga: 7091666.666666667
}
{
  _id: 'Aksesoris',
  rata_rata_harga: 3150000
}
{
  _id: 'Minimarket',
  rata_rata_harga: 58000
}

```

7. No 7

```
> db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        name: "Smart Watch",
        price: 3500000,
        stock: 10
      }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

8. No 8

```
> db.toko.aggregate([
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      harga_tertinggi: {
        $max: "$products.price"
      }
    }
  },
  {
    $sort: {
      harga_tertinggi: -1
    }
  }
])
< {
  _id: 'Surabaya',
  harga_tertinggi: 12000000
}
{
  _id: 'Bandung',
  harga_tertinggi: 8000000
}
{
  _id: 'Jakarta',
  harga_tertinggi: 7700000
}
{
```

```
{  
  _id: 'Purwokerto',  
  harga_tertinggi: 78000  
}
```

9. No 9

```
> db.toko.aggregate([
  {
    $match: {
      category: "Elektronik"
    }
  },
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      total_stock: {
        $sum: "$products.stock"
      }
    }
  },
  {
    $sort: {
      total_stock: -1
    }
  }
])
< {
  _id: 'Bandung',
  total_stock: 33
}
{
  _id: 'Surabaya',
  total_stock: 29
}
```

```
{  
  _id: 'Jakarta',  
  total_stock: 18  
}
```

10. No 10

```
> db.toko.deleteOne({  
  $and: [  
    { category: "Fashion" },  
    { city: "Surabaya" }  
  ]  
})  
< {  
  acknowledged: true,  
  deletedCount: 0  
}
```